

LAB EXPERIMENT RECORD

SET 7: CUSTOMER DEMOGRAPHICS ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Date	3 October 2026
Day	Saturday

Aim

To analyze customer demographic data using a bar chart, a pie chart and a demographic table in R, and to build an interactive dashboard using Tableau.

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

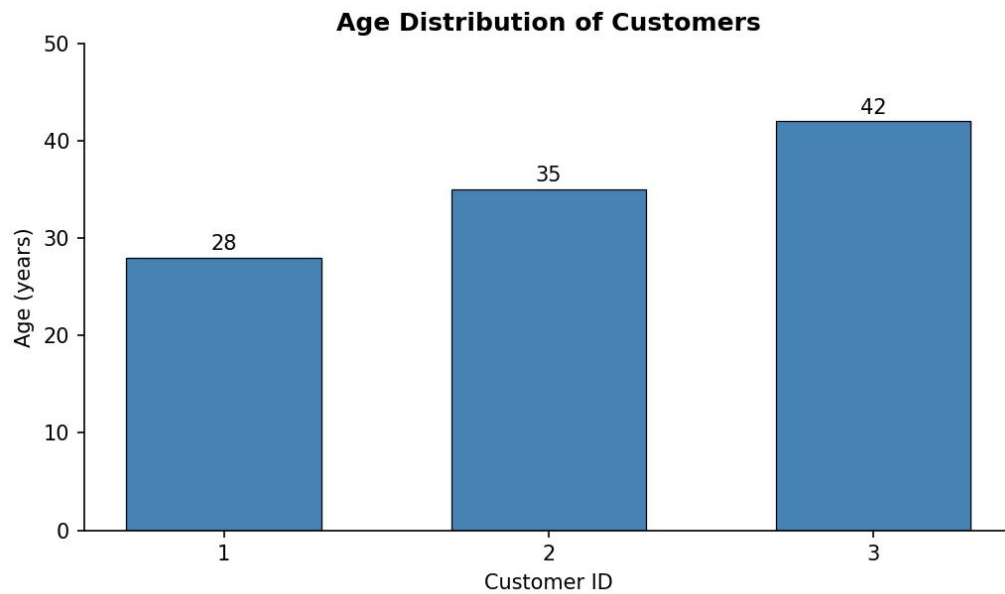
PROGRAM 1: Bar Chart of Customer Ages

Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

```
# Dataset
customer_id <- c(1, 2, 3)
age          <- c(28, 35, 42)

# Bar Chart
bp <- barplot(age,
               names.arg = customer_id,
               xlab = "Customer ID",
               ylab = "Age (years)",
               main = "Age Distribution of Customers",
               col = "steelblue",
               ylim = c(0, 50))

# Show the age value above each bar
text(bp, age, labels = age, pos = 3)
```



Output 1: Age Distribution of Customers (Bar Chart)

Observation: Customer 3 is the oldest (42 years) and Customer 1 is the youngest (28 years). The three ages are evenly spread, each 7 years apart, and the average customer age is 35 years.

PROGRAM 2: Pie Chart of Customers by Gender

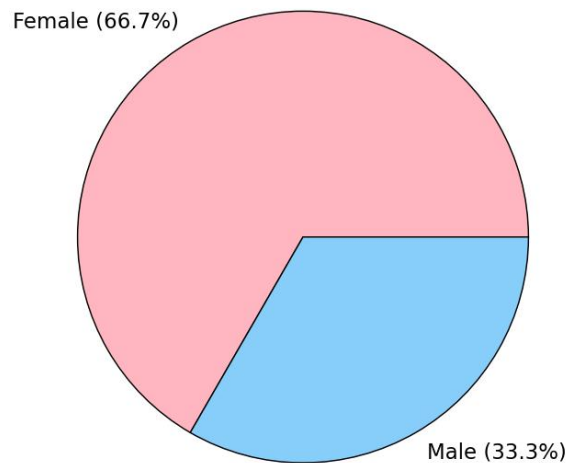
Generate a pie chart to represent the distribution of customers by gender.

```
# Dataset
gender <- c("Female", "Male", "Female")

# Count and percentage of customers in each gender
gender_count <- table(gender)
percent      <- round(gender_count / sum(gender_count) * 100, 1)
labels       <- paste0(names(gender_count), " (", percent, "%)")

# Pie Chart
pie(gender_count,
    labels = labels,
    col    = c("lightpink", "lightskyblue"),
    main   = "Distribution of Customers by Gender")
```

Distribution of Customers by Gender



Output 2: Distribution of Customers by Gender (Pie Chart)

Observation: Two of the three customers are female (66.7%) and one is male (33.3%), so female customers form the larger part of the customer group.

PROGRAM 3: Table of Customer Demographic Information

Build a table to show the demographic information for each customer. Label the table elements.

```
# Dataset with labelled columns
customers <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age         = c(28, 35, 42),
  Gender      = c("Female", "Male", "Female"),
  Income      = c(50000, 60000, 75000)
)

# Display the table
print(customers)

# Average age and average income
mean(customers$Age)
mean(customers$Income)
```

Console Output

```
Customer_ID Age Gender Income
1          1  28 Female  50000
2          2  35   Male  60000
3          3  42 Female  75000

[1] 35
[1] 61666.67
```

Output 3a: Customer Demographic Table (R Console)

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50,000
2	35	Male	60,000
3	42	Female	75,000

Output 3b: Customer Demographic Table with Labelled Columns

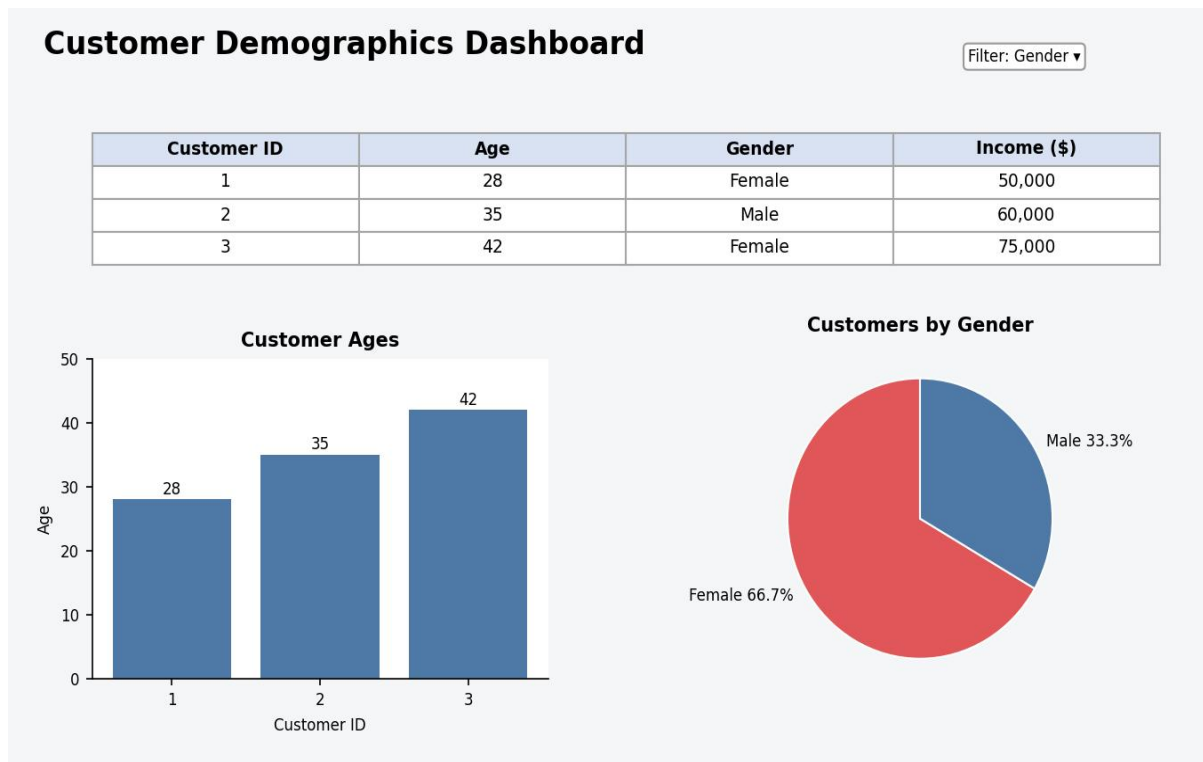
Observation: The table gives the age, gender and income of each customer. Income rises with age: Customer 1 (28 years) earns \$50,000, Customer 2 (35 years) earns \$60,000 and Customer 3 (42 years) earns \$75,000, which is the highest income. The average age is 35 years and the average income is about \$61,667.

PROGRAM 4: Interactive Customer Demographics Dashboard in Tableau

Develop a Tableau dashboard combining the bar chart, pie chart and the table for interactive exploration of customer demographics.

Steps in Tableau

1. Save the data (Customer ID, Age, Gender, Income) as an Excel or CSV file and connect to it in Tableau (Connect > Microsoft Excel / Text file). Set Customer ID to Dimension (discrete) so it is not summed.
2. Sheet 1 (Bar Chart): Drag Customer ID to Columns and Age to Rows (SUM). Add Age to Label to show the value above each bar. Rename the sheet "Customer Ages".
3. Sheet 2 (Pie Chart): Change the Marks type to Pie. Drag Gender to Color, Number of Records (or COUNT of Customer ID) to Angle, and add Number of Records to Label with a Percent of Total quick table calculation. Rename the sheet "Gender Share".
4. Sheet 3 (Table): Drag Customer ID and Gender to Rows, and drag Measure Names to Columns, keeping only Age and Income. Drag Measure Values to Text. Rename the sheet "Customer Table".
5. Create a new Dashboard (Dashboard > New Dashboard), set the size to Automatic, and drag the three sheets onto the dashboard area.
6. Add interactivity: select each chart and click Use as Filter, so clicking a bar or a pie slice filters the other views. Add a Gender filter (Show Filter > Apply to Worksheets > All Using This Data Source).
7. Add the title "Customer Demographics Dashboard" and test the filters in Presentation Mode.



Output 4: Interactive Customer Demographics Dashboard (Tableau)

Observation: The dashboard shows the customer ages, the gender distribution and the demographic table together. Clicking a pie slice (for example, Female) or a bar updates the other views, so users can explore the customers by gender and by age.

Procedure

1. Define the customer ID, age, gender and income values in R as vectors and a data frame.
2. Use the `barplot()` function to create a bar chart of customer ages, with axis labels and a title.
3. Use `table()` to count the customers by gender, calculate the percentages, and draw the pie chart with `pie()`.
4. Create a data frame with labelled columns, display it with `print()`, and calculate the average age and income with `mean()`.
5. Load the customer file into Tableau and build the bar chart, pie chart and table worksheets.
6. Combine the worksheets in a Tableau dashboard and add filters and actions for interactivity.

RESULT:

The customer demographic data was successfully analyzed using a bar chart, a pie chart and a demographic table in R. An interactive Tableau dashboard was also created, combining all three views so that users can explore the customer demographics by age and by gender.

Key Functions and Tools Used

Function / Tool	Purpose
<code>c()</code> / <code>data.frame()</code>	Creates the customer vectors and the demographic table.
<code>barplot()</code> / <code>text()</code>	Creates the bar chart and adds value labels.
<code>table()</code> / <code>paste0()</code>	Counts customers by gender and builds the pie chart labels.
<code>pie()</code>	Creates the pie chart.
<code>print()</code> / <code>mean()</code>	Displays the table and calculates the average age and income.
Tableau Marks (Pie)	Converts a worksheet into a pie chart.
Tableau Dashboard	Combines worksheets with filters and actions.